Sahaj Bhadja

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Summary

Product-minded software engineer experienced in **React, Node.js, TypeScript, PostgreSQL, scalable APIs, AI products, testing, and monitoring**, shipping reliable full-stack features for financial, enterprise, and data-intensive workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built FastAPI services and AI workflows processing 5,000+ monthly jobs with explicit state transitions, failure isolation, and validation, increasing concurrent throughput 3x while achieving 99%+ production reliability.
- Designed multi-tenant PostgreSQL and SQL Server schemas with row-level security for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through indexing and query optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Shipped React and Flask product workflows integrating multimodal AI models, OCR, AWS RDS, and REST APIs, serving 150+ instructors and reliably processing 10,000+ daily requests at scale.
- Implemented concurrent secure ingestion for 1,000+ files per batch, adding automated tests, failure handling, diagnostic logs, and fallback paths while reliably reducing feedback turnaround time by 60%.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*

- Developed Python and PyTorch components processing 100,000+ trajectory records, improving long-horizon planning 18% through rapid prototyping, simulation, debugging, reproducible testing, and iterative feature delivery at production scale.
- Collaborated with senior researchers to evaluate edge cases and model interactions, increasing closed-loop stability 15% while documenting technical findings, experimental tradeoffs, and reproducible implementation decisions at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed cross-platform Java and Python microservices with REST APIs, AWS Glue, S3, SQL, and Docker, improving low-bandwidth response times 30% and reducing payload sizes 25% at scale.
- Owned API and ETL improvements from requirements through deployment, increasing pipeline efficiency 68%, reducing technical debt 20%, and collaborating with Microsoft and GitHub stakeholders during Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built React and Node.js product features across eight microservices, synchronizing two million records through REST APIs, WebSockets, Azure SQL, authentication, permissions, notifications, and billing workflows at scale.
- Automated testing and Docker-based staged deployments through Jenkins, reducing release time 50% while using monitoring insights and code reviews to improve secure, maintainable full-stack delivery at scale.

Skills

Frontend: React, TypeScript, JavaScript, Next.js, HTML, CSS, TailwindCSS
Backend: Node.js, Express.js, Python, Java, FastAPI, Flask, REST APIs, WebSockets, microservices
Data: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, SQL, data modeling, query optimization
Cloud: AWS, GCP, Azure, Docker, Kubernetes, S3, CI/CD, Jenkins, GitHub Actions
Quality: Git, Jest, Mocha, PyTest, Mockito, unit testing, integration testing, monitoring, Agile

Projects

FinNexus Live | *TypeScript, Next.js, Financial Technology*

- Built a fintech platform supporting 50,000+ users and one million transactions with TypeScript, Next.js, MongoDB, Plaid bank linking, secure token controls, real-time updates, and responsive advanced filtering.

Big-Brain Live | *AI SaaS, PostgreSQL, GCP*

- Engineered AI-powered retrieval over 50,000+ documents using PostgreSQL, GCP, REST APIs, FAISS, and Pinecone, achieving sub-second search while reducing retrieval latency 40% through performance tuning at scale.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Developed an AI-powered SQL workspace across Java and TypeScript services, supporting 10+ databases with schema-aware query generation, structured validation, Docker workflows, and cross-platform local deployment at scale.